LECTURE 7

Function Limits, Continuity, and the Topology of Continuous Maps

MATH S4061 | Introduction to Modern Analysis I

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Natura non facit saltus.

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CONTENTS OF THIS LECTURE

Overview. This lecture begins Chapter 4 by replacing the sequential limit $p_n \to p$ with the function limit $\lim_{x\to p} f(x) = q$ in metric spaces. The ε–δ definition is tied back to sequences, then specialized to continuity at a point and continuity on a set. With the open-set characterization in hand, continuity becomes a topological operation: preimages of open sets are open, compositions are continuous, compact sets have compact images, connected sets have connected images, and the classical extreme-value and intermediate-value theorems become consequences. The lecture closes with homeomorphisms and the compactness theorem that upgrades pointwise continuity to uniform continuity. The companion text is Rudin, Chapter 4, especially R 4.1–R 4.26.

7.1 Function Limits

Theorem 7.1.1

Extreme values on a closed interval

If $f : [a, b] \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum: there exist $x, y \in [a, b]$ such that

$$f(x) \le f(z) \le f(y) \qquad \text{for every } z \in [a, b].$$

This is the interval case of Rudin's extreme-value theorem (R 4.16); the proof is deferred until compactness of images is available in 7.5.2.

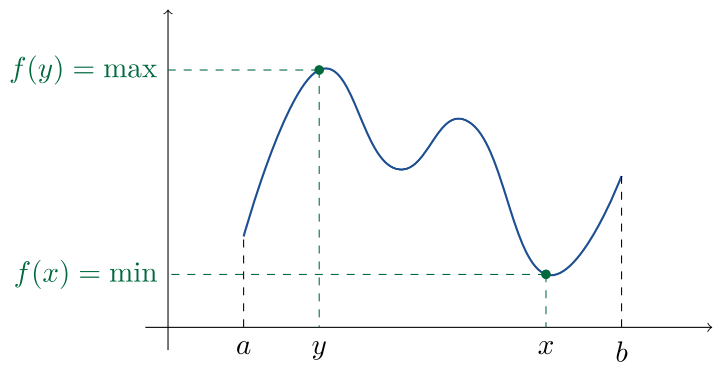


Figure 7.1.2. THE CALCULUS PICTURE OF EXTREMA. A continuous real-valued function on a closed interval cannot miss its highest and lowest values. The rigorous proof will come from compactness of the image (7.5.2).

Remark 7.1.3

Sequences as functions

For a sequence $\{p_n\}$ in a metric space X, writing $p_n \to p$ means $\lim_{n\to\infty} p_n = p$ in the ε–N sense. A sequence is simply a function

$$f : \mathbb{N} \to X, \qquad n \mapsto p_n,$$

and asks what happens as n gets large (cf. R 2.7, R 3.1).

Definition 7.1.4

Function limit in metric spaces

Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$, let p be a cluster point of E, and let $f : E \to Y$. We say that the *limit of f as $x \to p$ is q* (R 4.1) if for every $\varepsilon > 0$ there exists $\delta_{p,\varepsilon} > 0$ such that

$$0 < d_X(x, p) < \delta_{p,\varepsilon} \quad \Longrightarrow \quad d_Y\big(f(x), q\big) < \varepsilon.$$

We write

$$f(x) \to q \quad \text{as } x \to p, \qquad \text{or} \qquad \lim_{x\to p} f(x) = q.$$

The condition $0 < d_X(x, p)$ forces $x \neq p$; therefore $f(p)$ need not be defined, and even when it is defined its value does not affect this limit.

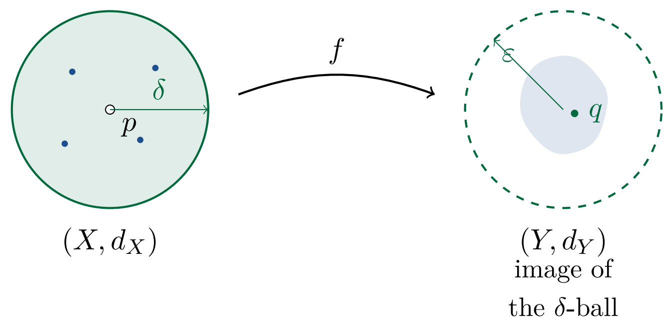


Figure 7.1.5. THE ε–δ PICTURE. The punctured δ-ball around p in the domain is carried inside the ε-ball around q in the target.

Proposition 7.1.6

Sequential criterion for function limits

Let (X, d_X) and (Y, d_Y) be metric spaces, let $E \subseteq X$, let $f : E \to Y$, and let p be a cluster point of E. Then

$$\lim_{x \to p} f(x) = q \quad \Longleftrightarrow \quad f(x_n) \to q \text{ for every sequence } \{x_n\} \subseteq E \setminus \{p\} \text{ with } x_n \to p.$$

This is Rudin's sequential characterization of function limits (R 4.2).

Proof 7.1.6

Proof by the ε–δ Definition and Its Contrapositive.

(1) Suppose $\lim_{x \to p} f(x) = q$ and let $\{x_n\} \subseteq E \setminus \{p\}$ satisfy $x_n \to p$. Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < d_X(x, p) < \delta \quad \Longrightarrow \quad d_Y(f(x), q) < \varepsilon.$$

Since $x_n \to p$, for all large n one has $d_X(x_n, p) < \delta$, and the punctured condition is automatic because $x_n \neq p$. Hence $d_Y(f(x_n), q) < \varepsilon$ eventually, so $f(x_n) \to q$.

(2) Conversely, prove the contrapositive. If the function limit fails, then for some $\varepsilon_0 > 0$ every $\delta > 0$ fails. Taking $\delta = 1/n$, choose $x_n \in E$ with

$$0 < d_X(x_n, p) < \frac{1}{n} \qquad \text{but} \qquad d_Y(f(x_n), q) \geq \varepsilon_0.$$

Then $x_n \to p$, while $f(x_n)$ does not converge to q. This contradicts the sequential condition, so the function limit holds.

■

[A2+A11] cf. R 4.2; use $\delta_n = 1/n$ in the reverse direction.

The logic is: if A is the function-limit statement and B is the sequential statement, prove $A \Rightarrow B$ and $\neg A \Rightarrow \neg B$.

7.2 Continuity and Composition

Definition 7.2.1

Continuity at a point

Let $f : E \to Y$ and let $p \in E$. The function f is *continuous at* p (R 4.5) if

$$\lim_{x \to p} f(x) = f(p).$$

Equivalently, for every $\varepsilon > 0$ there exists $\delta_{\varepsilon,p} > 0$ such that

$$d_X(x, p) < \delta_{\varepsilon,p} \quad \Longrightarrow \quad d_Y\big(f(x), f(p)\big) < \varepsilon.$$

For continuity the ball around p is not punctured: the point $x = p$ is allowed, and the limit value is required to be exactly $f(p)$ (cf. R 4.6 when p is a cluster point).

Definition 7.2.2

Continuity on a set

A function $f : E \to Y$ is *continuous on* E if it is continuous at every $p \in E$ (R 4.5).

Remark 7.2.3

Isolated points

If p is an isolated point of E, then f is continuous at p vacuously: for a sufficiently small ball, there are no points of E near p except p itself (cf. R 4.5).

Proposition 7.2.4

Composition of continuous functions

If $f : X \to Y$ is continuous at x and $g : Y \to Z$ is continuous at $f(x)$, then $g \circ f : X \to Z$ is continuous at x. Consequently, the composition of continuous functions is continuous (R 4.7).

Proof 7.2.4

Proof by an ε–η–δ Chase.

(1) Let $\varepsilon > 0$. Since g is continuous at $f(x)$, there exists $\eta > 0$ such that

$$d_Y(y, f(x)) < \eta \quad \Longrightarrow \quad d_Z\big(g(y), g(f(x))\big) < \varepsilon.$$

(2) Since f is continuous at x, there exists $\delta > 0$ such that

$$d_X(u, x) < \delta \quad \Longrightarrow \quad d_Y\big(f(u), f(x)\big) < \eta.$$

(3) Combining the two implications, $d_X(u, x) < \delta$ implies

$$d_Z((g \circ f)(u), (g \circ f)(x)) < \varepsilon,$$

so $g \circ f$ is continuous at x.

■

[A3] cf. R 4.7.

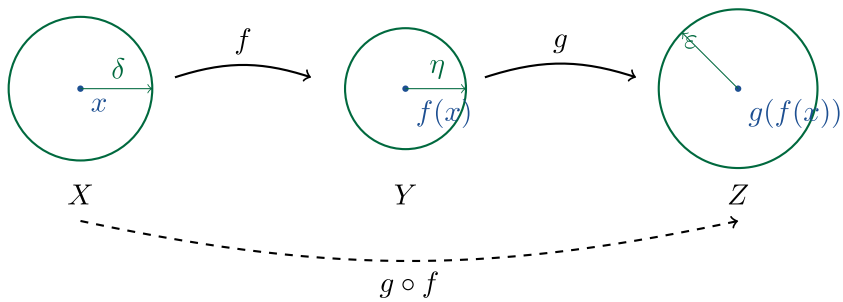


Figure 7.2.5. The composition chase. Continuity of g supplies the η-ball in Y; continuity of f supplies the δ-ball in X whose image lands inside it.

Remark 7.2.6

Sequential proof of composition

The sequential criterion gives the same result at a glance:

$$x_n \to x \implies f(x_n) \to f(x) \implies g(f(x_n)) \to g(f(x)).$$

This is the sequential version of R 4.7, using R 4.2.

7.3 Examples and Discontinuities

Example 7.3.1

Basic continuous real functions

The constant function $f(x) \equiv c$ is continuous: given $\varepsilon > 0$, every value of f already lies in $(c-\varepsilon, c+\varepsilon)$, so any $\delta > 0$ works. The identity $f(x) = x$ is continuous by taking $\delta = \varepsilon$. Building from constants and the identity through the algebra of limits and composition, every polynomial is continuous, and every rational function $p(x)/q(x)$ is continuous where $q(x) \neq 0$ (cf. R 4.9–R 4.11).

Proposition 7.3.2

The reciprocal is continuous on $(0, \infty)$

The function $f(t) = 1/t$ is continuous on $(0, \infty)$, but the required δ depends on both the point x and ε. This is a concrete ε–δ proof of the reciprocal case covered abstractly by R 4.9.

Proof 7.3.2

Direct ε–δ Proof for the Reciprocal.

(1) Fix $x > 0$ and $\varepsilon > 0$. If $|t - x| < \delta$, then

$$\left|\frac{1}{t} - \frac{1}{x}\right| = \frac{|t-x|}{tx}.$$

(2) First keep t away from 0. If $\delta \le x/2$, then $t > x/2$, so $tx > x^2/2$ and

$$\left|\frac{1}{t} - \frac{1}{x}\right| < \frac{\delta}{x^2/2} = \frac{2\delta}{x^2}.$$

(3) Therefore it is enough to require $\delta < \varepsilon x^2/2$. One may take

$$\delta(x, \varepsilon) = \min\left(\frac{x}{2}, \frac{\varepsilon x^2}{2}\right).$$

This proves continuity at x. The factor x^2 shows why δ shrinks as $x \to 0$. ∎

[A3] cf. R 4.9; the point-dependence is the point of the example.

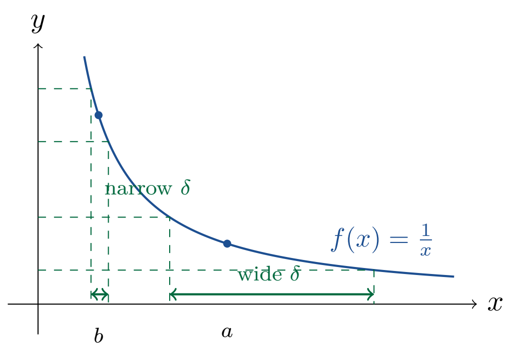


Figure 7.3.3. POINT-DEPENDENT δ FOR $1/x$. The same ε-band in the target pulls back to a narrow input interval where the graph is steep and a wider interval where the graph is flat.

Remark 7.3.4

Target and domain roles

The ε always measures closeness in the target Y; the δ always measures closeness in the domain X. In 7.3.2 the roles do not swap. Only the size of δ changes with the base point (cf. R 4.5–R 4.8).

Remark 7.3.5

Power series and familiar functions

Power series define continuous functions inside their radius of convergence. In particular,

$$e^x = \sum_{n \ge 0} \frac{x^n}{n!}$$

is continuous, and similarly $\sin x$, $\cos x$, $\tan x$, and the other familiar functions are continuous on their natural domains. Compare the radius-of-convergence discussion in 6.6.11 and Rudin's power-series radius theorem R 3.39.

Example 7.3.6

Standard discontinuities

The floor function $\lfloor x \rfloor$ is discontinuous at each integer. The piecewise function

$$f(x) = \begin{cases} e^x, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

is discontinuous at 0, since $\lim_{x\to 0^+} f(x) = 1$ while $\lim_{x\to 0^-} f(x) = 0$. Compare Rudin's catalogue of discontinuity examples in R 4.27.

Definition 7.3.7

Type I and Type II discontinuities

For a real function discontinuous at x_0, the one-sided limits classify the discontinuity.

- Type I: both one-sided limits exist, but either they differ (a jump) or they agree without matching $f(x_0)$ (a removable discontinuity).
- Type II: at least one one-sided limit fails to exist. This is often called an essential discontinuity.

This matches Rudin's first-kind/second-kind terminology (R 4.25–R 4.26).

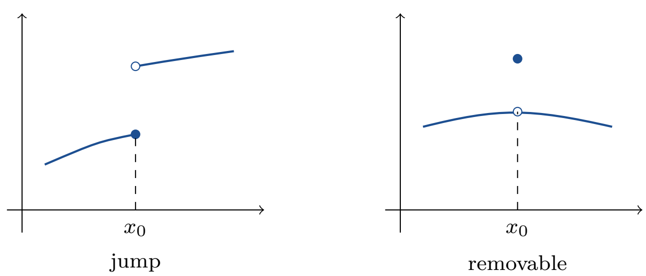


Figure 7.3.8. Type I discontinuities. A jump has different one-sided limits; a removable discontinuity has a hole at the limiting value and the function value placed elsewhere.

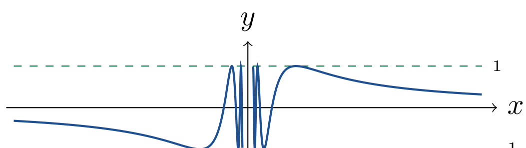
Example 7.3.9

The essential discontinuity of $\sin(1/x)$

The function

$$f(x) = \begin{cases} \sin(1/x), & x \neq 0, \\ 0, & x = 0, \end{cases}$$

has no limit at 0. Near 0 the graph oscillates infinitely often between -1 and 1; for different sequences $x_n \to 0$, the values $f(x_n)$ can approach different limits (cf. R 4.27).



$f(x) = \sin(1/x)$ near 0

Figure 7.3.10. A TYPE II DISCONTINUITY. The graph of $\sin(1/x)$ oscillates indefinitely near 0, so no single limiting value can govern all approaches to 0.

7.4 Open-Set Characterizations

Proposition 7.4.1

Sequential characterization of continuity

A function $f : E \to Y$ is continuous at $p \in E$ if and only if for every sequence $\{x_n\} \subseteq E$ with $x_n \to p$, one has $f(x_n) \to f(p)$. This specializes Rudin's sequential limit criterion to continuity (cf. R 4.2, R 4.6).

Proof 7.4.1

Proof by Specializing the Limit Criterion.

(1) If f is continuous at p, apply 7.1.6 with $q = f(p)$; the non-punctured case causes no issue, since a sequence equal to p along a subsequence already has image equal to $f(p)$ along that subsequence.

(2) Conversely, if continuity fails at p, then for some $\varepsilon_0 > 0$ every $\delta > 0$ fails. With $\delta = 1/n$, choose $x_n \in E$ such that

$$d_X(x_n, p) < \frac{1}{n} \qquad \text{but} \qquad d_Y(f(x_n), f(p)) \geq \varepsilon_0.$$

Then $x_n \to p$, but $f(x_n)$ does not converge to $f(p)$.

∎

[A2+A11] cf. R 4.2; the reverse direction again uses $\delta_n = 1/n$.

Proposition 7.4.2

Continuity at a point via neighbourhoods

The function $f : X \to Y$ is continuous at p if and only if for every open set $V \subseteq Y$ containing $f(p)$, there is an open set $U \subseteq X$ containing p such that

$$f(U) \subseteq V \qquad \text{equivalently} \qquad U \subseteq f^{-1}(V).$$

This is the pointwise neighbourhood form underlying Rudin's global open-set characterization (R 4.8).

Proof 7.4.2

Proof by Translating Balls into Open Neighbourhoods.

(1) If f is continuous at p and $V \subseteq Y$ is open with $f(p) \in V$, choose $\varepsilon > 0$ such that $B_\varepsilon(f(p)) \subseteq V$. Continuity gives $\delta > 0$ such that

$$f(B_\delta(p)) \subseteq B_\varepsilon(f(p)) \subseteq V.$$

Taking $U = B_\delta(p)$ gives the required neighbourhood.

(2) Conversely, take $V = B_\varepsilon(f(p))$. The neighbourhood condition gives an open $U \ni p$ with $f(U) \subseteq V$. Since U is open, some $B_\delta(p) \subseteq U$, and hence

$$f(B_\delta(p)) \subseteq B_\varepsilon(f(p)).$$

This is exactly the ε–δ condition for continuity at p.

■

[A3]

Proposition 7.4.3

Global open-set characterization of continuity

A function $f : X \to Y$ is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set $V \subseteq Y$. This is Rudin's open-set characterization of continuity (R 4.8).

Proof 7.4.3

Proof by Pulling Back Open Balls.

(1) Suppose f is continuous and let $V \subseteq Y$ be open. To show $f^{-1}(V)$ is open, take $x \in f^{-1}(V)$. Since V is open, there is $\varepsilon > 0$ with $B_\varepsilon(f(x)) \subseteq V$. Continuity at x gives $\delta > 0$ with

$$f(B_\delta(x)) \subseteq B_\varepsilon(f(x)) \subseteq V.$$

Hence $B_\delta(x) \subseteq f^{-1}(V)$, so every point of $f^{-1}(V)$ is interior.

(2) Conversely, fix $x \in X$ and $\varepsilon > 0$. The ball $B_\varepsilon(f(x))$ is open in Y, so $f^{-1}(B_\varepsilon(f(x)))$ is open in X by hypothesis. Since x lies in this preimage, there is $\delta > 0$ with

$$B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x))).$$

Thus $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$, so f is continuous at x. Since x was arbitrary, f is continuous on X.

■

[Direct] cf. R 4.8.

Proposition 7.4.4

Composition via preimages

The composition of continuous functions is continuous. This is a second proof of R 4.7, using R 4.8.

Proof 7.4.4

Proof by the Open-Set Characterization.

(1) Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ with f and g continuous, and let $V \subseteq Z$ be open.

(2) Since g is continuous, $g^{-1}(V)$ is open in Y. Since f is continuous, $f^{-1}(g^{-1}(V))$ is open in X.

(3) But preimage commutes with composition:

$$(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V)).$$

Thus the preimage under $g \circ f$ of every open set is open, so $g \circ f$ is continuous.

■

[Direct] a second proof of 7.2.4; cf. R 4.8.

This is the clean topological proof of composition: it works because preimages, not images, commute exactly with composition.

Remark 7.4.5

Pugh's moral

The sequential criterion is often the easiest way to check continuity at a glance: to test continuity at p, push every sequence $x_n \to p$ through the function and see whether $f(x_n) \to f(p)$ (cf. R 4.2).

7.5 Continuous Images of Compact and Connected Sets

Theorem 7.5.1

Continuous image of a compact set

If $f : X \to Y$ is continuous and X is compact, then the image

$$f(X) = \{y \in Y : \exists\, x \in X \text{ with } f(x) = y\}$$

is compact in Y (R 4.14).

Proof 7.5.1

Proof by Pulling Back an Open Cover.

(1) Let $\{V_\alpha\}$ be an open cover of $f(X)$, with each V_α open in Y. Since f is continuous, each $f^{-1}(V_\alpha)$ is open in X.

(2) These preimages cover X:

$$X = \bigcup_\alpha f^{-1}(V_\alpha).$$

Indeed, if $x \in X$, then $f(x) \in f(X)$, so $f(x) \in V_\alpha$ for some α, and hence $x \in f^{-1}(V_\alpha)$.

(3) By compactness of X, choose finitely many indices $\alpha_1, \ldots, \alpha_n$ such that

$$X = f^{-1}(V_{\alpha_1}) \cup \cdots \cup f^{-1}(V_{\alpha_n}).$$

Applying f, the sets $V_{\alpha_1}, \ldots, V_{\alpha_n}$ cover $f(X)$. Thus every open cover of $f(X)$ has a finite subcover, so $f(X)$ is compact.

■

[A6] cf. R 4.14; uses 7.4.3.

Corollary 7.5.2

Extreme value theorem

If $f : [a, b] \to \mathbb{R}$ is continuous, then it attains a maximum and a minimum: there exist $m, M \in [a, b]$ such that

$$f(m) \leq f(x) \leq f(M) \qquad \text{for all } x \in [a, b].$$

This is Rudin's extreme-value theorem for real-valued continuous functions on compact domains (R 4.16).

Proof 7.5.2

Proof by Compactness of the Image.

(1) The interval $[a, b]$ is compact, so $f([a, b])$ is compact by 7.5.1. By Heine–Borel in $\mathbb{R}$, it is closed and bounded.

(2) Since $f([a, b])$ is bounded, the real numbers

$$\alpha = \inf f([a, b]), \qquad \beta = \sup f([a, b])$$

exist.

(3) Since $f([a, b])$ is closed, it contains its infimum and supremum. Thus $\alpha, \beta \in f([a, b])$, so there are $m, M \in [a, b]$ with

$$f(m) = \alpha, \qquad f(M) = \beta.$$

Therefore $f(m) \leq f(x) \leq f(M)$ for every $x \in [a, b]$.

■

[A6+A5] cf. R 4.16; uses 7.5.1 and Heine–Borel.

Definition 7.5.3

Connected metric space

A metric space (X, d) is *connected* if it cannot be written as $X = A \cup B$ where

$$A \neq \emptyset, \qquad B \neq \emptyset, \qquad A \cap B = \emptyset, \qquad A, B \text{ are open.}$$

Such a decomposition is called a *separation*. Equivalently, the only clopen subsets of X are $\emptyset$ and X itself. In $\mathbb{R}$, the connected subsets are exactly the intervals (cf. R 2.45, R 2.47).

Theorem 7.5.4

Continuous image of a connected set

If X is connected and $f : X \to Y$ is continuous, then $f(X)$ is connected (R 4.22).

Proof 7.5.4

Proof by Pulling Back a Separation.

(1) Suppose for contradiction that $f(X) = A \cup B$ is a separation of $f(X)$: the sets A and B are nonempty, disjoint, and open in the subspace $f(X)$.

(2) Then $f^{-1}(A)$ and $f^{-1}(B)$ cover X, are disjoint, and are open in X by continuity. They are also nonempty: because $A, B \subseteq f(X)$ are nonempty, each contains some value $f(x)$.

(3) Thus $X = f^{-1}(A) \cup f^{-1}(B)$ is a separation of X, contradicting connectedness. Therefore $f(X)$ is connected.

■

[A9] cf. R 4.22; uses 7.4.3.

Corollary 7.5.5

Intermediate value theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous and assume, without loss of generality, that $f(a) \le f(b)$. Then for every $y \in [f(a), f(b)]$ there exists $x \in [a, b]$ with $f(x) = y$. This is the intermediate-value theorem (R 4.23).

Proof 7.5.5

Proof by Connectedness of the Image.

(1) The interval $[a, b]$ is connected, so $f([a, b])$ is connected by 7.5.4. Since connected subsets of $\mathbb{R}$ are intervals, $f([a, b])$ is an interval.

(2) This interval contains $f(a)$ and $f(b)$, and therefore contains every $y \in [f(a), f(b)]$. Hence for each such y there is $x \in [a, b]$ with $f(x) = y$.

■

[Direct] cf. R 4.23; compactness gives closed and bounded; connectedness gives an interval.

The structural mirror is useful: continuity preserves compactness, giving maxima and minima; continuity preserves connectedness, giving intermediate values.

7.6 Homeomorphisms

Two metric spaces are considered topologically the same when there is a bijection between them that preserves the open-set structure in both directions.

Definition 7.6.1

Homeomorphism

A *homeomorphism* is a continuous bijection $f : X \to Y$ whose inverse $f^{-1} : Y \to X$ is also continuous. If such an f exists, then X and Y are *homeomorphic*. Rudin uses the compact-domain theorem for continuous bijections in R 4.17.

Theorem 7.6.2

A continuous bijection from a compact domain is a homeomorphism

Let $f : X \to Y$ be continuous and bijective. If X is compact, then $f^{-1} : Y \to X$ is continuous, so f is a homeomorphism (R 4.17).

Proof 7.6.2

Proof by Showing the Inverse Pulls Back Opens to Opens.

(1) To prove f^{-1} is continuous, it is enough to show that for every open $V \subseteq X$, the preimage of V under f^{-1} is open in Y. Since f is a bijection,

$$(f^{-1})^{-1}(V) = f(V).$$

(2) Let $V \subseteq X$ be open. Then V^c is closed in X, and since X is compact, V^c is compact.

(3) By 7.5.1, $f(V^c)$ is compact in Y, hence closed. Because f is a bijection,

$$f(V) = \big(f(V^c)\big)^c.$$

Therefore $f(V)$ is open in Y. Thus f^{-1} is continuous.

■

[A6] cf. R 4.17; compactness makes continuous images of closed sets closed.

Counterexample 7.6.3

A continuous bijection need not be a homeomorphism

Let

$$\varphi : [0, 2\pi) \to S^1 \subseteq \mathbb{R}^2, \qquad \varphi(t) = (\cos t, \sin t).$$

This map is a continuous bijection onto the unit circle, but $[0, 2\pi)$ is not compact and φ^{-1} is not continuous: points on S^1 approaching $(1, 0)$ from the fourth quadrant have arguments tending to 2π, while $\varphi^{-1}(1, 0) = 0$ (cf. R 4.21).

FAILS: the claim that every continuous bijection is automatically a homeomorphism.

Remark 7.6.4

The compactness engine

The proof of 7.6.2 says that f^{-1} is continuous exactly when f carries open sets to open sets; equivalently, it is enough here that f carry closed sets to closed sets. On a compact domain this is automatic:

closed subsets of X are compact, continuous images of compact sets are compact, and compact subsets of a metric space are closed (cf. R 4.17).

Remark 7.6.5

Recognising homeomorphic spaces

The basic homeomorphism questions are: given metric spaces M and N, are they homeomorphic, and what are the continuous maps $M \to N$? Typical comparison examples include the unit circle, a trefoil knot, the perimeter of a square, the 2-torus, $[0, 1]$, and the unit disc. Compare the compact-domain homeomorphism theorem R 4.17 and circle example R 4.21.

7.7 Uniform Continuity

Definition 7.7.1

Uniform continuity

A function $f : X \to Y$ is *uniformly continuous* on X (R 4.18) if for every $\varepsilon > 0$ there exists $\delta_\varepsilon > 0$ such that

$$d_X(p, q) < \delta_\varepsilon \quad \Longrightarrow \quad d_Y\big(f(p), f(q)\big) < \varepsilon.$$

The difference from ordinary continuity is that δ depends only on ε, not on the base point.

Theorem 7.7.2

Compactness gives uniform continuity

If X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous (R 4.19).

Proof 7.7.2

Proof by a Finite Half-Radius Cover.

(1) Let $\varepsilon > 0$. Since f is continuous at each $p \in X$, choose $\delta(p) > 0$ such that

$$d_X(p, q) < \delta(p) \quad \Longrightarrow \quad d_Y\big(f(p), f(q)\big) < \frac{\varepsilon}{2}.$$

(2) Define the half-radius ball

$$J(p) := B^X_{\delta(p)/2}(p).$$

The collection $\{J(p)\}_{p \in X}$ is an open cover of X, so compactness gives $p_1, \ldots, p_N$ with

$$X \subseteq J(p_1) \cup \cdots \cup J(p_N).$$

(3) Set

$$\delta = \frac{1}{2} \min(\delta(p_1), \ldots, \delta(p_N)) > 0.$$

This δ depends only on ε.

(4) Take $x, y \in X$ with $d_X(x, y) < \delta$. Since the $J(p_i)$ cover X, choose i with $x \in J(p_i)$.

Then

$$d_X(p_i, x) < \frac{1}{2}\delta(p_i),$$

so

$$d_Y\big(f(p_i), f(x)\big) < \frac{\varepsilon}{2}.$$

(5) Also,

$$d_X(y, p_i) \le d_X(y, x) + d_X(x, p_i) < \delta + \frac{1}{2}\delta(p_i) \le \delta(p_i),$$

so

$$d_Y\big(f(y), f(p_i)\big) < \frac{\varepsilon}{2}.$$

(6) By the triangle inequality in Y,

$$d_Y\big(f(y), f(x)\big) \le d_Y\big(f(y), f(p_i)\big) + d_Y\big(f(p_i), f(x)\big) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus f is uniformly continuous.

■

[A6+A4] cf. R 4.19; the half-radius leaves room for nearby points.

Counterexample 7.7.3

The reciprocal is not uniformly continuous on $(0, \infty)$

The function $f(x) = 1/x$ is continuous on $(0, \infty)$, but not uniformly continuous there. In the explicit continuity estimate from 7.3.2,

$$\delta(x, \varepsilon) = \min\left(\frac{x}{2}, \frac{\varepsilon x^2}{2}\right) \to 0 \qquad \text{as } x \to 0.$$

Thus no single δ works for all $x > 0$. The domain $(0, \infty)$ is not compact, so 7.7.2 does not apply. On a compact interval $[a, b]$ with $a > 0$, the same function is uniformly continuous. Compare Rudin's noncompact-domain counterexamples in R 4.20.

FAILS: the claim that ordinary continuity on a noncompact domain implies uniform continuity.

Remark 7.7.4

Three compactness-for-free results

Compactness upgrades continuity in three recurring ways: a continuous image of a compact set is compact (7.5.1); a continuous bijection from a compact domain is a homeomorphism (7.6.2); and a continuous map from a compact domain is uniformly continuous (7.7.2). These are R 4.14, R 4.17, and R 4.19.

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